

Exploratory Data Analysis of the Disease Symptoms and Patient Profile Dataset

1. Introduction

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, distribution, and characteristics of the Disease Symptoms and Patient Profile Dataset before applying machine-learning algorithms.

The dataset contains patient symptoms and profile information such as fever, cough, fatigue, difficulty breathing, age, gender, blood pressure, and cholesterol level.

The target variable is Disease, which contains multiple disease categories. Therefore, the project is treated as a supervised multi-class classification problem.

The EDA helps identify missing values, duplicate records, disease-class distribution, feature distributions, numerical characteristics, and potential outliers before applying Multinomial Logistic Regression.

2. Dataset Overview

The original dataset contains 349 patient records and 10 columns.

Dataset Details

Property	Value
Total Records	349
Total Columns	10
Unique Diseases	116
Numerical Features	1
Categorical Columns	9
Target Variable	Disease
Learning Type	Supervised Learning

Columns in the Dataset

1. Disease
2. Fever
3. Cough
4. Fatigue
5. Difficulty Breathing
6. Age
7. Gender
8. Blood Pressure
9. Cholesterol Level
10. Outcome Variable

The first few records contain patient information such as Influenza, Common Cold, Eczema, and Asthma along with their symptoms and profile characteristics.

Observation

The dataset contains 349 records representing 116 different disease classes. The large number of disease categories makes the problem a multi-class classification task.

3. Dataset Structure and Data Types

The `df.info()` analysis shows that all 349 records contain values for every column. The dataset has 1 numerical column (Age) and 9 object/categorical columns.

Column	Data Type	Role
Disease	Object / Categorical	Target
Fever	Object / Categorical	Input Feature
Cough	Object / Categorical	Input Feature
Fatigue	Object / Categorical	Input Feature
Difficulty Breathing	Object / Categorical	Input Feature
Age	Integer	Input Feature
Gender	Object / Categorical	Input Feature

Column	Data Type	Role
Blood Pressure	Object / Categorical	Input Feature
Cholesterol Level	Object / Categorical	Input Feature
Outcome Variable	Object / Categorical	Additional Dataset Column

All columns contain 349 non-null values.

Observation

The dataset contains both numerical and categorical information. Age is the only numerical feature, while symptoms and patient profile characteristics are categorical.

4. Missing Value Analysis

Missing values were checked using `df.isnull().sum()`.

Column	Missing Values
Disease	0
Fever	0
Cough	0
Fatigue	0
Difficulty Breathing	0
Age	0
Gender	0
Blood Pressure	0
Cholesterol Level	0
Outcome Variable	0

Total Missing Values: 0

The notebook confirms that every column contains 0 missing values.

Observation

The dataset is complete and does not require missing-value imputation before further analysis.

5. Duplicate Data Analysis

Duplicate records were checked using the `duplicated()` function.

Result

- Total records = 349
- Duplicate records = 49
- Unique records after removing duplicates = 300
- Duplicate percentage $\approx 14.04\%$

The notebook subsequently removed duplicate records using `drop_duplicates()`, changing the dataset shape from (349, 10) to (300, 10).

Observation

The dataset contains 49 duplicate records, which account for approximately 14.04% of the original dataset.

Removing duplicate records prevents repeated observations from influencing the analysis and machine-learning model.

6. Target Variable Analysis

The target variable is Disease.

The dataset contains 116 unique disease classes.

Top Disease Classes

Disease	Number of Records	Percentage
Asthma	23	6.59%
Stroke	16	4.58%
Osteoporosis	14	4.01%
Diabetes	10	2.87%

Disease	Number of Records	Percentage
Hypertension	10	2.87%

The remaining disease classes occur with smaller frequencies, with several diseases having only one record.

Observation

The target variable has 116 disease classes, but the number of records is not equally distributed among the classes.

For example, Asthma has 23 records, whereas several disease classes have only 1 record. This indicates a significant class-imbalance problem.

This is important because very small classes provide limited examples for training a multi-class classification model.

7. Feature Distribution Analysis

The categorical features were analyzed using count plots to understand the distribution of patient characteristics.

7.1 Gender Distribution

Gender	No.of Patients
Female	176
Male	173
Total	349

Observation

The gender distribution is almost balanced, with 176 female and 173 male patients.

7.2 Fever Distribution

Fever	No.of Patients
Yes	175
No	174

Fever	No.of Patients
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Total	349
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Observation

The distribution of fever is almost equal, with 175 patients having fever and 174 patients not having fever.

7.3 Cough Distribution

Cough	Number of Patients
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No	182
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Yes	167
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Total	349
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Observation

A slightly larger number of patients, 182, do not have cough, while 167 patients have cough.

7.4 Fatigue Distribution

Fatigue	Number of Patients
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Yes	242
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No	107
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Total	349
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Observation

Fatigue is present in 242 patients, while 107 patients do not have fatigue.

Therefore, fatigue is one of the more frequently occurring symptoms in the dataset.

7.5 Difficulty Breathing Distribution

Difficulty Breathing	Number of Patients
Yes	88
No	261
Total	349

Observation

Most patients, 261, do not have difficulty breathing, while 88 patients have difficulty breathing.

7.6 Blood Pressure Distribution

Blood Pressure	Number of Patients
Low	18
Normal	164
High	167
Total	349

Observation

Most patients have either High blood pressure (167) or Normal blood pressure (164). Only 18 patients have low blood pressure.

7.7 Cholesterol Level Distribution

Cholesterol Level	Number of Patients
Low	149
Normal	34
High	166

Cholesterol Level	Number of Patients
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Total	349
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Observation

High cholesterol is the most frequent category with 166 patients, followed by low cholesterol with 149 patients. Only 34 patients have normal cholesterol levels.

7.8 Outcome Variable Distribution

Outcome	Number of Patients
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Positive	186
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Negative	163
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Total	349
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Observation

The dataset contains 186 positive and 163 negative outcome records.

However, the Outcome Variable is not used as an input feature in the final disease prediction model. The notebook removes this column before separating the features and target.

8. Numerical Summary

The numerical analysis was performed on the Age feature using descriptive statistics.

Statistic	Age
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Count	349
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Mean	46.32
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Std Deviation	13.09
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Minimum	19
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Q1 (25%)	35
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Median	45
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Statistic	Age
Q3 (75%)	55
Maximum	90

Observation

The average patient age is approximately 46.32 years, while the median age is 45 years.

The ages range from 19 to 90 years, showing that the dataset represents patients from young adults to elderly individuals.

The mean and median are relatively close, indicating that the overall age distribution is not extremely skewed.

9. Outlier Analysis

Outlier analysis was performed on the Age feature using the Interquartile Range (IQR) method.

Calculations

- $Q1 = 35$
- $Q3 = 55$
- $IQR = Q3 - Q1$
- $IQR = 55 - 35 = 20$

Lower Bound

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$= 35 - 1.5 \times 20$$

$$= 5$$

Upper Bound

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

$$= 55 + 1.5 \times 20$$

$$= 85$$

Therefore:

- Lower Bound = 5 years
- Upper Bound = 85 years
- Potential outliers = 2

- Outlier values = 90 years

Observation

Two records with an age of 90 years were identified as potential outliers because they are above the upper IQR boundary of 85.

These records were not removed, because an age of 90 years can represent a valid patient.

10. Data Preprocessing After EDA

After the EDA, duplicate records were removed.

The dataset changed from:

349 records → 300 records

The Outcome Variable column was then removed because it was not selected as an input feature for disease prediction.

Input Features

The final input features used for the model are:

1. Fever
2. Cough
3. Fatigue
4. Difficulty Breathing
5. Age
6. Gender
7. Blood Pressure
8. Cholesterol Level

Target

Disease

Categorical features were converted into numerical representations using one-hot encoding. The encoded features included:

- Age
- Fever_Yes
- Cough_Yes
- Fatigue_Yes

- Difficulty Breathing_Yes
 - Gender_Male
 - Blood Pressure_Low
 - Blood Pressure_Normal
 - Cholesterol Level_Low
 - Cholesterol Level_Normal
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11. Visualization Analysis

Several visualizations were generated during EDA to understand the dataset.

11.1 Top 20 Disease Classes

A bar chart was created using the frequency of disease classes.

The graph shows that Asthma has the highest number of records with 23, followed by Stroke with 16 and Osteoporosis with 14.

11.2 Age Distribution

A histogram with KDE was used to visualize the distribution of patient ages.

The age values range from 19 to 90 years, with a mean of 46.32 years.

11.3 Categorical Feature Distributions

Count plots were generated for:

- Gender
- Fever
- Cough
- Fatigue
- Difficulty Breathing
- Blood Pressure
- Cholesterol Level
- Outcome Variable

These plots show the frequency of each category and help identify dominant categories in the dataset.

11.4 Age Boxplot / Outlier Analysis

The age distribution was examined using the IQR method. Two potential outliers with age 90 were identified.

12. Overall EDA Findings

EDA Component	Finding
Total Records	349
Total Columns	10
Unique Diseases	116
Missing Values	0
Duplicate Records	49
Records After Removing Duplicates	300
Numerical Feature	Age
Age Mean	46.32
Age Median	45
Age Minimum	19
Age Maximum	90
Potential Age Outliers	2
Most Frequent Disease	Asthma – 23
Second Most Frequent Disease	Stroke – 16
Third Most Frequent Disease	Osteoporosis – 14
Most Common Gender	Female – 176
Most Common Symptom	Fatigue – Yes, 242
Most Common Blood Pressure	High – 167
Most Common Cholesterol Level	High – 166
Outcome Positive	186

EDA Component	Finding
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Outcome Negative	163
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The EDA shows that the dataset is complete with no missing values, but it contains 49 duplicate records and a highly distributed multi-class target with 116 diseases.

13. Conclusion

The Exploratory Data Analysis of the Disease Symptoms and Patient Profile Dataset provided an understanding of the dataset structure, feature distributions, target-variable distribution, and numerical characteristics.

The dataset initially contains 349 records and 10 columns, with no missing values. However, 49 duplicate records were identified and removed, leaving 300 records for further processing.

The target variable Disease contains 116 classes, with Asthma being the most frequent disease with 23 records. The analysis also showed that fatigue is present in 242 patients, while the age of patients ranges from 19 to 90 years.

Two potential age outliers were identified at 90 years, but they were retained because they can represent valid patient ages.

After EDA, the duplicate records were removed, the Outcome Variable was excluded, categorical features were encoded, and the resulting features were prepared for Multinomial Logistic Regression.

Overall, the EDA provides a clear understanding of the dataset and establishes the foundation for the subsequent multi-class disease prediction using Multinomial Logistic Regression.